Summary of Module 2: Health Data Sharing

This summary synthesizes the core concepts of Module 2, focusing on the regulatory landscape for health data sharing in the European Union. The primary lecture materials for this module are `Health data sharing pt 1.md` and `Health data sharing pt 2.md`. This document expands upon the lecture content by integrating detailed analysis from supplementary papers and the legal text of the regulations themselves, particularly the European Health Data Space (EHDS) Regulation.

1. The European Data Strategy and the Rise of Data Spaces

The European Union's overarching data strategy aims to create a single market for data, allowing it to flow freely within the EU and across sectors for the benefit of businesses, researchers, and public administrations. A central pillar of this strategy is the creation of common European data spaces in strategic sectors, including health. The **European Health Data Space (EHDS)** is the first and most prominent of these proposed data spaces, designed to address the unique challenges and opportunities of health data.

The fundamental goal of the EHDS is to balance two key objectives: 1. **Empowering Individuals**: Giving natural persons greater access to and control over their personal health data. 2. **Fostering a Single Market**: Creating a consistent and efficient framework for the use and reuse of health data for research, innovation, policy-making, and regulatory activities.

This is detailed in Recital 1 of the EHDS Regulation (`Module_2/OJ_L_202500327_EN_TXT.md`). The COVID-19 pandemic served as a major catalyst, highlighting the critical need for timely access to quality health data for public health surveillance and crisis response (Recital 2, EHDS Regulation).

2. The Layered Legal Framework for Data Sharing

The EHDS does not exist in a vacuum. It is built upon and interacts with a multi-layered legal framework governing data in the EU. Understanding this hierarchy is crucial.

2.1. The Foundation: General Data Protection Regulation (GDPR)

The GDPR is the bedrock of data protection in the EU. For health data, it establishes that "data concerning health" is a special category of personal data (Article 9, GDPR), requiring a specific legal basis for processing.

A key right under the GDPR is the **right to data portability** (Article 20). However, as the lectures and the ITRE committee study (`Module_2/IPOL_STU(2022)740054_EN.md`) point out, this right has significant limitations in the health context: - **Limited Scope**: It primarily

applies to data "provided by the data subject," not to "inferred data" such as a doctor's diagnosis. - **Technical Feasibility**: The right to have data transmitted directly from one controller to another is only guaranteed "where technically feasible," but the GDPR does not mandate the creation of such technical feasibility.

The EHDS is designed to overcome these specific limitations, acting as a *lex specialis* (a more specific law) for health data.

2.2. The Horizontal Pillars: DGA and Data Act

Two horizontal regulations create a broader framework for data sharing that the EHDS fits into:

- **Data Governance Act (DGA)**: This regulation aims to increase trust in data sharing. As explained in `Health data sharing pt 1.md`, its key contributions are establishing rules for **data intermediation services** (neutral third parties that facilitate data sharing) and a framework for **data altruism** (individuals and companies consenting to the use of their data for the common good). The EHDS builds on this by setting specific rules for data altruism in the health sector (Recital 78, EHDS Regulation).
- **Data Act (DA)**: This regulation focuses on fairness in the digital economy. Its most relevant provision is granting users of connected devices (IoT) the right to access and share the data generated by their use of those products or services. In the health context, this applies to data from medical equipment and wellness devices. The EHDS complements the Data Act by specifying rules for this data, for instance clarifying that no compensation is required for sharing data for primary use (Recital 24, `IPOL_STU(2022)740054_EN.md`).

2.3. The Sector-Specific Regulation: The European Health Data Space (EHDS)

The EHDS Regulation is the central piece of legislation for this module. It creates a specific, harmonized framework for health data, distinguishing between two fundamental types of use.

Primary Use: Data for Healthcare Primary use is defined as the processing of electronic health data for the provision of healthcare to an individual (Article 2(d), EHDS Regulation). The goal is to ensure continuity of care, especially across borders.

Key Provisions for Primary Use: - Strengthened Patient Rights: The EHDS goes beyond the GDPR by granting natural persons the right to **immediate, free access** to their personal electronic health data, including inferred data like diagnoses (Article 3, EHDS Regulation). They also gain the right

to have this data transmitted to a provider of their choice in a common, interoperable format. - **MyHealth@EU**: This is the mandatory cross-border infrastructure for the primary use of health data. It connects national contact points, allowing, for example, a pharmacy in Spain to dispense an electronic prescription issued in Finland (Article 23, EHDS Regulation). Participation, which was previously voluntary under the Cross-Border Healthcare Directive, becomes compulsory. - **Mandatory Interoperability**: The regulation mandates that Electronic Health Record (EHR) systems comply with common standards and specifications to ensure they can "speak the same language." This is a direct response to the fragmentation that has hindered data sharing to date (Recital 7, EHDS Regulation).

Secondary Use: Data for Research, Innovation, and Policy Secondary use involves using health data for broader societal benefits, such as scientific research, developing new medicines, or informing public health policy (Article 2(e), EHDS Regulation).

Key Provisions for Secondary Use: - Health Data Access Bodies (HDABs): Each Member State must establish one or more HDABs. These bodies act as trusted gatekeepers, providing a single point of access for researchers and other data users. They are responsible for assessing data access applications and issuing "data permits" (Article 64, EHDS Regulation). - **Data Permit System**: A health data user (e.g., a university researcher) must apply to an HDAB for a data permit, specifying the purpose of their request. The HDAB assesses whether the purpose is legitimate, as defined in the regulation (Article 34), and not among the prohibited uses (Article 35). - **Secure Processing Environments**: A core principle of secondary use is that the data does not move to the researcher; the researcher comes to the data. Data is made available in a highly secure processing environment, and only anonymised or aggregated results can be downloaded. Direct access to raw, pseudonymised data is strictly controlled within this environment to protect patient privacy (Article 77, EHDS Regulation). - **HealthData@EU**: This is the cross-border infrastructure for secondary use. It connects the national HDABs, allowing a researcher to request data from multiple Member States through a single application, in principle (Article 80, EHDS Regulation).

3. Key Concepts and Debates

The lecture materials and supplementary readings highlight several critical concepts and ongoing debates surrounding the EHDS.

3.1. Patient Rights: Access, Control, and the Opt-Out Mechanism

The EHDS significantly strengthens patient rights. However, it also introduces a crucial debate around patient control.

- **Right to Restrict Access (Primary Use)**: Natural persons have the

right to restrict access to parts of their health data (e.g., sensitive information). However, they must be warned that this could impact their care, and in life-threatening emergencies, this restriction can be overridden to protect their vital interests (Article 8 & 11, EHDS Regulation).

- **Right to Opt-Out (Secondary Use)**: For secondary use, natural persons have a right to opt out of having their data made available. This is a key point of contention. The critical policy paper, "The European health data space: Too big to succeed?" (`Module_2/1-s2.0-S016885102300146X-main.md`), argues that a high opt-out rate could undermine the very purpose of the EHDS by creating incomplete and biased datasets, rendering them useless for reliable research. The regulation attempts to balance this by allowing Member States to override the opt-out for research of "important reasons of public interest," but this remains a sensitive and controversial area (Recital 54, EHDS Regulation).

3.2. The Role and Burden of Health Data Access Bodies (HDABs)

The HDABs are the linchpin of the secondary use framework. However, their success is not guaranteed. The ITRE study (`Module_2/IPOL_STU(2022)740054_EN.md`) raises concerns that the proposal may be "too centralised" and "place too great a burden on the national Health Data Access Body." The list of duties for HDABs is extensive, including processing applications, preparing datasets, ensuring security, and monitoring compliance. There is a significant risk that these bodies could become bottlenecks if not adequately resourced and efficiently managed.

3.3. Economic and Societal Value vs. Privacy

The economic justification for the EHDS is substantial. The ITRE study notes that the benefits of primary use (e.g., reducing information asymmetries and medical errors) and secondary use (e.g., accelerating drug development) likely far exceed the implementation costs.

However, this value is predicated on the availability of vast amounts of high-quality data. The critical policy paper (`Module_2/1-s2.0-S016885102300146X-main.md`) frames the central challenge as a "trilemma" between **data quantity**, **data quality**, and **citizen trust**. Maximizing data quantity and quality for researchers may come at the cost of citizen trust if data protection safeguards are perceived as weak. Conversely, overly strict privacy controls (like a widely used opt-out) could decimate the data pool. The success of the EHDS will depend on its ability to navigate this trilemma effectively.

3.4. Prohibited Uses and Unintended Consequences

Article 35 of the EHDS Regulation lists prohibited secondary uses, such as using data to increase insurance premiums or for advertising. While designed to

protect citizens, the ITRE study suggests these prohibitions may be too rigid and could have unintended consequences. For example: - **Insurance**: Prohibiting any use by insurance companies might prevent them from designing more efficient and outcome-based payment models for healthcare providers, which could ultimately benefit patients. - **Marketing**: A blanket ban on marketing use might also prevent valuable market research that could lead to better, more competitive pharmaceutical products.

The study recommends a more nuanced approach, perhaps focusing on prohibiting *harmful* outcomes rather than entire categories of use.

4. Conclusion

Module 2 presents the European Health Data Space as a landmark legislative effort to unlock the potential of health data. It is built on a complex, multi-layered legal framework and aims to strike a delicate balance between individual rights, public health, and economic innovation.

The EHDS proposes a sophisticated architecture with `MyHealth@EU` for personal healthcare and `HealthData@EU` with its network of `Health Data Access Bodies` for research and policy. While the potential benefits are immense, its success hinges on overcoming significant challenges, including ensuring robust data quality, managing the administrative burden on new institutions, and, most importantly, maintaining the trust of EU citizens. The debates over the opt-out mechanism and the precise scope of permitted data uses will be central to its implementation and ultimate impact.